Exp No: 18
Date : 08/10/26

a) Implement echo client server using TCP/UDP Socket

AIM:

To develop a python-based client-server communication system using TCP/UDP sockets and threading, where the client sends message to the server and the server receives & responds to those message.

TCP Program:

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET, socket.
                                        SOCET_STREAM)

    server.bind('127.0.0.1', 5000)
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")
    conn, addr = server.accept()
    print(f"[SERVER] connected by {addr}")
    while True:
        data = conn.recv(1024).decode()
            if not data or data.lower() == 'quit':
                print("[SERVER] connection closed by
                                        client")
                break
            print(f"[SERVER] Received: {data}")
            conn.send(f"server received: {data}".encode())
            conn.close()
            server.close()
```

```
def start_client():
    client = socket.socket(socket.AF_INET, socket.
                                    SOCKET_STREAM)

    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")
    while True:
        msg = input("Enter message or 'quit' to exit):
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
        client.close()

if __name__ == "__main__":
    Server_thread = threading.Thread(target = start_server
                                    daemon = True)

    Server_thread.start()
    start_client()
```

O/P:

[SERVER] Received : hello

[SRRVER] RESPONSE] Serva received : Hello.

Enter a message (or 'quit' to exit): quit.

UDP:

```
import socket
import threading

def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 1200))
    print('[SERVER] Running on 127.0.0.1:1200")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
        print(f"[SERVER] client {addr} ended the
connection.")
        break
        print(f"[SERVER] Received from {addr}:
                {msg}")
        s.sendto(data, addr)
    s.close()
    print" [SERVER] Stopped.")

def udp_client():
    c = socket.socket(socket.AF_INET), socket.
                        SOCK_DGRAM)

    Server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), Server)
```

```
if msg.lower() == 'quit':
    print("[CLIENT] closing connection..")
    break

data, _ = c.recvfrom(1024)
print("Echo from Server:", data.decode())
c.close()
print("[CLIENT] Stopped.")

if __name__ == "__main__":
    threading.Thread(target = udp_server, daemon=True).sta
    rt() udp_client()
```

O/P:

```
[Server] Received from ('127.0.0.1', 62859): Hello
Echo from Server : Hello
Enter message (or 'quit' to stop): quit
```